

DANIEL MORK, PH.D.

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EXPERIENCE

Independent Consultant	2021 - present
Harvard T.H. Chan School of Public Health	Boston, Massachusetts
Research Scientist, Department of Biostatistics	Jan 2024 - present
Research Associate, Department of Biostatistics	Aug 2022 - Dec 2023
Postdoctoral Research Fellow, Department of Biostatistics	Aug 2021 - Jul 2022
Colorado State University	Fort Collins, Colorado
Graduate Research Assistant	May 2018 - Aug 2021
Graduate Teaching Assistant	Aug 2017 - Dec 2019
Graybill Statistics and Data Science Laboratory	Fort Collins, Colorado
Statistical Consultant	May 2019 - Dec 2020
Greeley-Evans School District 6	Greeley, Colorado
Mathematics Teacher, Greeley West High School	Aug 2013 - May 2017
Student Council Advisor, Greeley West High School	Aug 2013 - May 2017
Target Corporation	Minneapolis, Minnesota
Logistics, Reporting, & Analytics; In-Store Marketing	Oct 2011 - Apr 2013
No-Boundaries Web Development and Marketing	
Owner/Operator	2005 - 2011

EDUCATION

Colorado State University , Fort Collins, CO	2017 - 2021
Ph.D. Statistics, <i>Thesis: Bayesian treed distributed lag models, Advisor: Ander Wilson</i>	
M.S. Statistics (2019)	
University of Saint Thomas , St. Paul, MN	2011 - 2013
M.A. Mathematics Education	
Certificate, Engineering Education	
Saint Olaf College , Northfield, MN	2007 - 2011
B.A. (<i>Cum Laude</i>) Mathematics, Psychology, Norwegian	
Concentration, Nordic Studies	

PUBLICATIONS

* student mentored † co-first author

1. *Dinehart C, Delaney S, Mock L, Racette B, Miller B, Kioumourtzoglou M-A, Braun D, Zanobetti A, **Mork D** (2025). Extreme heat and hospitalization among older adults with Parkinson's Disease. *Journal of Exposure Science and Environmental Epidemiology*, revision invited.
2. Biswas A, Qiu M, Braun D, Dominici F, **Mork D** (2025). Quantifying the impact of solar energy on CO2 emissions from fossil fuel power plants: Regional variations and lagged effects. *Science Advances*, 11(31). <https://doi.org/10.1126/sciadv.adq5660>
3. **Mork D**, Wilson A (2025). Incorporating prior information into distributed lag nonlinear models with zero-inflated monotone regression trees. *Bayesian Analysis* 20(2): 627-655. <https://doi.org/10.1214/23-BA1412>
4. **Mork D**, Strawderman R, Audirac M, Dominici F, Ertefaie A (2025). Structural nested cumulative recurrent event model for estimating the effects of time-varying exposure on recurrent event outcomes. *Journal of the American Statistical Association*, in review. <https://arxiv.org/abs/2506.07910>
5. *McBrien H, **Mork D**, Kioumourtzoglou M-A, Casey J (2025). Assessing potential sources of bias in measuring power outage exposure with simulations. *Journal of Exposure Science and Environmental Epidemiology* 9(4):p e403. <https://doi.org/10.1097/EE9.0000000000000403>
6. *Im S, Wilson A, **Mork D** (2025). Structured Bayesian Regression Tree Models for Estimating Distributed Lag Effects: The R Package dlmtree. *R Journal*, 17(1). <https://doi.org/10.32614/RJ-2025-007>
7. Jung YS, Choma E, Delaney S, **Mork D**, Audirac M, Braun D, Coull B, Nadeau K, Zanobetti A (2025). Extreme Heat and Hospital Admissions in Older Adults: A Small-Area Analysis in the Greater Boston Metropolitan Area. *Environmental Research*, 9(3), p.e395. <https://doi.org/10.1097/EE9.0000000000000395>
8. *Spoto F, Tian J, Hügel J, Ortega DT, Ritchie CS, Blacker D, Dominici F, Patel CJ, **Mork D**, Estiri H. Quantifying Diagnostic Signal Decay in Dementia: A National Study of Medicare Hospitalization Data. *In review*.
9. Mock L, **Mork D**, Audirac M, Braun D, Zanobetti A, Delaney S. Hospitalization risk with and without dementia by region and race in the US *Annals of Internal Medicine*, in review.
10. *Cork M, **Mork D**, Dominici F. Overcoming Local Confounding in Causal Inference for Continuous Treatments. *Annals of Applied Statistics*, to be submitted.
11. *Kitch J, Woodward S, Braun D, Bell M, Cork M, Zanobetti A, Dominici F, **Mork D**. Inferential impacts of spatial confounding in national-scale air pollution health analyses. *Journal of the American Statistical Association*, to be submitted.
12. *Spoto F, Delaney S, Braun D, Bell M, Zanobetti A, **Mork D**. Acute effects of heat and fine particulate matter on hospitalization among patients with Alzheimer's disease and related dementias. *Environmental Epidemiology*, in review.
13. *McBrien H, **Mork D**, Do V, Kioumourtzoglou M-A, Casey J. Power outages increase cardiovascular and respiratory hospitalizations among US older adults. *PNAS*, in review.
14. *Sarkar J, **Mork D**, Braun D, Aggarwal S, Nethery R, Parks R, Delaney S, Zanobetti A (2025). Flood Exposure and Neuropsychiatric Hospitalization among the Medicaid Population in the U.S., 2000-2012. *American Journal of Public Health*, in review.
15. *Im S, **Mork D**, Leung M, Weisskopf M, Kioumourtzoglou MA, Wilson A. Treed distributed lag mixture model with zero-inflated count data to investigate the association between air pollution and pregnancy loss. *Statistics in Medicine*, in review.
16. *Im S, **Mork D**, Wilson A. Heterogeneous Distributed Lag Mixture Model for Precision Environmental Health with Longitudinally Assessed Mixture Exposures. *Journal of the Royal Statistical Society: Series C*, in review.
17. *Qin MM, Khoshnevis N, Dominici F, Braun D, Zanobetti A, **Mork D** (2025). Comparing traditional and causal inference methodologies for evaluating impacts of long-term air pollution exposure on hospitalization with Alzheimer's disease and related dementia. *American Journal of Epidemiology* 194(1), pg 64-72. <https://doi.org/10.1093/aje/kwae133>

18. [†]Delaney S, ^{*†}Stegmuller A, **Mork D**, Bell M, Gill T, Braun D, Zanobetti A (2025). Extreme heat and hospitalization among older persons with Alzheimer's Disease and Related Dementias. *JAMA Internal Medicine*. <https://doi.org/10.1001/jamainternmed.2024.7719>
19. *Dong S, Braun D, Wu X, Yitshak-Sade M, Blacker D, Kioumourtzoglou M-A, **Mork D**, Schwartz J, Dominici F, Zanobetti A (2025). The impacts of air pollution on mortality and readmission among Medicare beneficiaries with Alzheimer's disease and Alzheimer's disease related dementias. *Lancet: Planetary Health* 9(2), pg e114-123. [https://doi.org/10.1016/S2542-5196\(25\)00001-4](https://doi.org/10.1016/S2542-5196(25)00001-4)
20. *Cork M, **Mork D**, Dominici F (2025). Methods for estimating the exposure-response curve to inform the new safety standards for fine particulate matter. *Journal of the Royal Statistical Society, Series A*. <https://doi.org/10.1093/jrssa/qnaf004> **Selected as JRSS Discussion Paper**
21. **Mork D**, Kioumourtzoglou MA, Weisskopf M, Coull BA, Wilson A (2024). Heterogeneous distributed lag models to estimate personalized effects of maternal exposures to air pollution. *Journal of the American Statistical Association*, 119(545), pg 14-26. <https://doi.org/10.1080/01621459.2023.2258595> **ASA Editor's Choice Collection**.
22. **Mork D**, Dominici F, Delaney S (2024). Policy-induced air pollution health disparities: Statistical and data science considerations. *Science*, 385, pg 391-396. <https://doi.org/10.1126/science.adp1870>
23. Wang VA, Delaney S, Flynn L, Racette BA, Miller GW, Braun D, Zanobetti A, **Mork D** (2024). The effect of air pollution on hospitalizations with Parkinson's disease among medicare beneficiaries nationwide. *npj Parkinson's Disease*, 10(196) <https://doi.org/10.1038/s41531-024-00815-x>
24. Klompmaker J, **Mork D**, Zanobetti A, Braun D, Hankey S, Hart J, Hystad P, Jimenez MP, Laden F, Larkin A, Lin PD, Suel E, Yi L, Zhang W, Delaney J, James P (2024). Associations of Street-View Greenspace with Parkinson's Disease Hospitalizations in Medicare. *Environment International* 188. <https://doi.org/10.1016/j.envint.2024.108739>
25. MacDougall H, **Mork D**, Hanson S, Henning-Smith C (2024). Rural-urban differences in health care unaffordability. *Journal of Rural Health* 40(2), pg 376-385. <https://doi.org/10.1111/jrh.12788>
26. Fann N, Zanobetti A, **Mork D**, Steinhardt W, Rappold A (2024). Applying a multi-state survival model to explore the role of fine particles in promoting frailty in the Medicare cohort. *Environmental Epidemiology* 8(1), pg 1-8. <https://doi.org/10.1097/EE9.0000000000000285>
27. Peat F, Kawcak C, McIlwraith W, Keenan D, Berk J, **Mork D** (2024). Subchondral lucencies of the medial femoral condyle in yearling and two-year-old Thoroughbred sales horses: Prevalence, progression and associations with racing performance. *Equine Veterinary Journal* 52(1), pg 99-109. <https://doi.org/10.1111/evj.13945>
28. Peat F, Kawcak C, McIlwraith W, Keenan D, Berk J, **Mork D** (2024). Radiological findings in the proximal sesamoid bones of yearling and two-year-old Thoroughbred sales horses: Prevalence, progression and associations with racing performance. *Equine Veterinary Journal* 57(1), 87-100. <https://doi.org/10.1111/evj.14051>
29. **Mork D**, Wilson A (2023). Estimating perinatal critical windows to environmental mixtures via structured Bayesian regression tree pairs. *Biometrics* 79(1), pg 449-461. <https://doi.org/10.1111/biom.13568>
30. ^{*†}Woodward S, [†]**Mork D**, Wu X, Zhewen H, Braun D, Dominici F (2023). Combining aggregate and individual-level data to estimate individual-level associations between air pollution and COVID-19 mortality in the United States. *PLOS Global Public Health* 3(8), p.e0002178. <https://doi.org/10.1371/journal.pgph.0002178>
31. **Mork D**, Braun D, Zanobetti A (2023). Time-lagged relationships between a decade of air pollution exposure and first hospitalization with Alzheimer's disease and related dementias. *Environment International* 171. <https://doi.org/10.1016/j.envint.2022.107694>
32. **Mork D**, Wilson A (2022). Treed distributed lag nonlinear models. *Biostatistics* 23(3), pg 754-771. <https://doi.org/10.1093/biostatistics/kxaa051>
33. Maitre L, et al, **The Exposome Data Challenge Participant Consortium [Mork D, et al]** (2022). State-of-the-art methods for exposure-health studies: Results from the exposome data challenge event. *Environment International* 168. <https://doi.org/10.1016/j.envint.2022.107422>
34. **Mork D** (2021). Bayesian treed distributed lag models. Doctoral dissertation, Colorado State University. <https://proquest.com/docview/2572884506>

CONFERENCE ABSTRACTS AND OTHER PUBLICATIONS

1. Louis C, Dimentberg R, Osho A, Olverson G, Bhagat R, Ziazadeh DR, **Mork D**, Coglianese EE, Givertz MM, Mehra MR, D'alessandro AD, Sundt TM, Itoh A (2025). Optimal Anticoagulation Intensity Limiting Thrombotic and Bleeding Outcomes Amongst Patients with 3rd Generation Left Ventricular Assist Device. *AATS*.
2. Delaney S, Klompmaker J, **Mork D**, Braun D, James P (2024). Differing associations between novel eye-level greenspace measures and Parkinsonian hospitalizations. *ISEE Conference Abstracts*.
3. Wang V, Delaney S, Braun D, Zanobetti A, **Mork D** (2024). Examining the effect of air pollutants on Parkinson's disease hospitalization among Medicare beneficiaries using traditional and causal inference approaches. *ISEE Conference Abstracts*.
4. Delaney S, Klompmaker J, **Mork D**, Braun D, Zanobetti A, James P (2024). Street-view green space impacts on Alzheimer's disease hospitalization risk in the US Medicare population. *ISEE Conference Abstracts*.
5. Delaney S, **Mork D**, Weisskopf M, Teimeier H, Braun D, Zanobetti A (2023). The impact of multiple marginalization on the association between acute exposure to hot summer temperature and reduced childhood cognitive performance. *ISEE Conference Abstracts*.
6. **Mork D**, Delaney S, Braun D, Zanobetti A (2023). The impacts of fine particulate matter components and trace elements on rates of dementia hospitalization. *ISEE Conference Abstracts*.
7. **Mork D**, Wilson A (2021). Estimating Perinatal Critical Windows to Environmental Mixtures via Structured Bayesian Regression Tree Pairs. *ISEE Conference Abstracts*.

FUNDING

Current

ECHO-wide precision environmental health analyses of air pollution exposure and birth outcomes. NIH/ECHO OIF award. PI: Mork.

Statistical Methods for Precision Environmental Health with Mixture Exposures. NIH/NIEHS R01. Role: Co-Investigator. PI: Wilson.

Statistical methods to characterize causal mechanisms by which air pollution affects the recurrence of cardiovascular events. NIH R01. Role: Co-Investigator. PI: Ertefaie/Strawderman/Dominici.

Characterizing the link between multiple environmental exposures and Parkinsons disease exacerbation. NIH/NIA R01 (09/20/2022 - 07/31/2027). Role: Researcher. PI: Zanobetti/Braun.

Susceptibility and adverse health outcomes related to climate-sensitive events among older Medicare beneficiaries with Alzheimer and Dementia. NIH/NIA RF1. Role: Researcher. PI: Bell/Dominici.

Completed

National cohort studies of Alzheimer's disease, related dementias and air pollution. NIH/NIA R01 (04/01/2020 - 01/31/2025). Role: Researcher. PI: Zanobetti/Braun/Dominici.

The confluence of extreme heat cold on the health and longevity of an Aging Population with Alzheimers and related Dementia. NIH/NIA RF1 (06/01/2022 - 05/31/2025). Role: Researcher. PI: Patel/Dominici/Estiri.

Air pollution, climate change, greenspace, and neurodegenerative diseases: in search of health equity-promoting strategies in Medicare. Harvard Chan-NIEHS/Biogen Pilot Project. Role: Co-Investigator. PI: Delaney/Zanobetti.

SOFTWARE FOR REPRODUCIBLE RESEARCH

R package: dlmtree

<https://github.com/danielmork/dlmtree>

Bayesian regression tree software for distributed lag methods written in C++, efficient procedures for posterior sampling. Includes methods for multiple publications:

- TDLNM, Mork & Wilson, Biostatistics
- TDLMM, Mork & Wilson, Biometrics
- HDLM, Mork et. al., JASA

- Monotone-TDLNM, Mork & Wilson, Bayesian Analysis

GitHub Repository: Air Pollution/ADRD/Distributed Lag

Code to create and analyze dataset for time-lagged pollution relationship with ADRD hospitalizations in a longitudinal study of over 27 million Medicare beneficiaries (Mork, Braun, Zanobetti, Environment International).

PRESENTATIONS

Invited

Society for Pediatric and Perinatal Epidemiology, <i>presentation</i>	2025
WNAR annual meeting, International Biometrics Society, <i>presentation</i>	2025
International Biometrics Conference, <i>presentation</i>	2024
Colorado University Anschutz, <i>seminar</i>	2024
WNAR annual meeting, International Biometrics Society, <i>presentation</i>	2024
Stanford Quantitative Sciences Unit, <i>presentation</i>	2023
Harvard T.H. Chan NIEHS Center Retreat, <i>poster</i>	2022
NSAPH seminar series, Harvard T.H. Chan School of Public Health, <i>presentation</i>	2022
Symposium, International Society for Environment Epidemiology, <i>presentation</i>	2021
Jane Warren award webinar, Health Effects Institute, <i>presentation</i>	2021
Post-doctoral job talk, Harvard T.H. Chan School of Public Health, <i>presentation</i>	2021
3 minute challenge, Colorado State University, Vice-President for Research (video)	2021

Contributed

ASA ENVR Workshop, <i>poster</i>	2024
NSAPH seminar series, Harvard T.H. Chan School of Public Health, <i>presentation</i>	2023
International Society for Environment Epidemiology annual conference, <i>presentation</i>	2023
ENAR spring meeting, International Biometrics Society, <i>presentation</i>	2022
Statistics department showcase, Colorado State University, <i>poster</i>	2021
Research talk, Environmental biostatistics working group, Colorado State Univ., <i>presentation</i>	2021
ENAR spring meeting, International Biometrics Society, <i>presentation</i>	2021
Graduate student showcase, Colorado State University, <i>poster & presentation</i>	2020
Joint statistical meetings (JSM), <i>presentation</i>	2020
Statistics department showcase, Colorado State University, <i>poster</i>	2020
Graduate student showcase, Colorado State University, <i>poster</i>	2019
Colorado conference of teachers of mathematics, Denver, CO, <i>presentation</i>	2014
Minnesota undergraduate psychology conference, Northfield, MN, <i>presentation</i>	2011
Student showcase, Saint Olaf College, <i>poster</i>	2010

HONORS/AWARDS

American Statistical Association Editor's Choice Collection	2023
Health Effects Institute, Jane Warren Trainee Conference Award	2021

Colorado State University

Vice president for research fellowship nominee/3-minute challenge participant	2021
College of natural sciences outstanding scholar award	2020
James R. zumBrunnen statistical collaboration award	2020
Statistics department poster competition: Best poster award	2020
Boes excellence in teaching award	2019

Greeley-Evans School District 6

District 6 star teacher	2015
Teacher of the month	2013

Other

ETS Recognition of excellence: Mathematics	2013
NSF scholarship (\$7,000/yr): Encouraging careers in mathematical sciences	2007 - 2011

TEACHING AND MENTORING

Harvard University

Graduate Dissertation/Thesis Advisor

Michael Cork, Biostatistics PhD, in progress	2021 -
JRSS Discussion Paper	

Shuxin Dong, Epidemiology PhD, in progress	2021 -
Jinia Sarkar, MPH	2023 - 2024
Claire Dinehart, MPH	2023 - 2024
Angela Stegmuller, MPH	2022 - 2023
Amit Kumar, MPH, <i>next: Scientist, Exponent, Bellevue WA</i>	2022 - 2023
<i>Undergraduate Thesis Advisor</i>	
Sarah Cao (thesis advisor),	2024 - 2025
Michelle Qin (thesis advisor), <i>next: Biostatistics PhD student at Johns Hopkins</i>	2022 - 2023
Sophie Woodward (thesis co-advisor), <i>next: Biostatistics PhD student at HSPH</i>	2021 - 2023
<i>General Supervision and Mentoring</i>	
James Kitch, Research Assistant, Biostatistics	2024 -
Federica Spoto, Postdoctoral Fellow, Biostatistics	2024 -
Veronica Wang, Postdoctoral Fellow, Environment Health	2023 -
Lauren Mock, Biostatistician	2023 -
Colorado State University	
<i>Courses</i>	
STAT 301: Introduction to applied statistical methods (Primary instructor)	F17, S18, F18
STAT 315: Introduction to theory and practice of statistics (Primary instructor)	S19, F20
STAT 556: Directed statistical consulting (Graduate assistant)	S20
<i>Workshops</i>	
Coding & Cookies Workshop: Data visualization using ggplot	F19, F20, S21
Coding & Cookies Video Series: Data visualization using ggplot	F20
Coding & Cookies Video Series: Tidy data in R	F20
Coding & Cookies Workshop: Data wrangling in R	F19, S20
Practical R on Summit high performance computing system	F18
<i>Graduate Research Mentoring</i>	
Seongwon Im	2021 -
<i>Graduate Statistical Consultant Mentoring</i>	
Connor Gibbs	2020
— <i>Project title: Machine learning methods for characterizing coffee beans using mass spectrometry data</i>	
Mantautaus Rimkus	2020
— <i>Project title: Structural equation modeling of longitudinal relationships between meaningful activities and meaning in life for U.S. military veterans in college</i>	
Emily Williams	2020
— <i>Project title: Relationships between sleep and activity habits of practicing physicians</i>	
<i>Graduate Instructor Mentoring</i>	
Bingying Dai, Troy Wixson, Wilson Wright	2020
Maddie Rainey, Gray Stanton, Zifeng Zhang	2019
Michael Creutzinger, Connor Gibbs, Hanxiao Jing, Mengting Lin	2018
<i>Undergraduate Instructor Mentoring</i>	
Caroline Thomas	2019
Greeley-Evans School District 6	
<i>Courses</i>	
Pre-Algebra; Algebra I and II; Geometry; Trigonometry; Pre-Calculus; IB Math HL	2013 - 2017
<i>Workshops</i>	
Personalized learning and standards-based grading: Theory and practice	2012
Curriculum planning and design	2011
<i>Probationary Teacher Mentoring</i>	
Scott Geels	2015

SERVICE & PROFESSIONAL MEMBERSHIP

<i>Service</i>	
ENAR Council for Emerging and New Statisticians (CENS)	

— Co-chair	2023 - 2024
— Steering committee	2022 - 2024
— Social media committee co-chair	2021 - 2022
ENAR Student paper competition review committee	2023, 2024
<i>Harvard T.H. Chan School of Public Health</i>	
Research Scientist Association Council	2024 -
National Studies on Air Pollution and Health research presentation series	2021, 2022, 2023
— Moderator, Discussant	
<i>Colorado State University</i>	
Teaching awards committee	2020
Prospective graduate student visit day panelist and representative	2018, 2019, 2020
Probability casino, assistant, Department of Statistics	2018
<i>Conference Organization</i>	
IMS New Researchers Conference, assistant organizer, Fort Collins, CO	2019
<i>Conference Session Organization</i>	
ENAR 2022: <i>Interdisciplinary engagement as a Biostatistician: Maximizing collaboration skills</i>	
ENAR 2023: <i>Kickstart Your Career: How to Maximize Those Early Years</i>	
ENAR 2024: <i>Paths to Success: Perspectives from Recruiters and Hiring Committees</i>	
WNAR 2024: <i>Advances in causal inference for environmental exposures</i>	
IBC 2024: <i>Empowering emerging biometricians: Developing an emerging biometrician group within IBS</i>	
WNAR 2025: <i>Innovative approaches to modeling the complex relationships between environmental mixtures and health</i>	
<i>Ad Hoc Reviewer</i>	
Annals of Applied Statistics	
Biometrics	
Biostatistics	
Data Science in Science	
Environmental Epidemiology	
Environmental Health Perspectives	
Environment International	
Environmetrics	
European Journal of Epidemiology	
International Journal of Public Health	
Journal of the Royal Statistical Society	
New England Journal of Medicine	
Scientific Reports	
Statistics and Computing	
Statistics in Medicine	
Statistical Methods in Medical Research	
<i>Grant Reviewer</i>	
MADRES Center for Environmental Health Disparities	
<i>Professional Membership</i>	
American Statistical Association	
International Biometrics Society	
The International Environmetrics Society	
International Society for Bayesian Analysis	
International Society for Environmental Epidemiology	
<i>Greeley-Evans School District 6</i>	
Innovation 2020: Personalized learning action team	2016 - 2017
Professional development committee	2014 - 2017
Mathematics curriculum vertical team	2014 - 2016
Hiring and interview team	2014 - 2016
Benchmark assessment committee	2014 - 2015